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================================================================================
                    Session 181    Creating a Conduit for Windows
================================================================================

With Developer Studio 97 (5.0) installed
With CDK 3.0 installed to
c:\cdk

================================================
To Create a Conduit Project
================================================
Developer Studio
File -> New...
New
  Projects tab
  select Palm Conduit Wizard(dll)
  Project name:
    mymemo
  Location:
    c:\cdk\samples\mymemo
  OK

Palm Conduit Wizard(dll) – Step 1
  What do you want the Conduit Wizard to generate?
    select Source code for the conduit
  Which App on the organizer will your conduit interface with?
    Application:  Palm Memo Pad
Next>
  What type of data transfer between the desktop and the organizer will your conduit provide?
    select Two way mirror image synchronization
Next>
  What features would you like?
    check Archiving
    check Sync action configuration dialog
    check File linking
Next>
  AppWizard creates the following classes for you:
   select the 1st one in the list
    Edit to
    Class name:          CMyMemoDll
    Header name:         MyMemo.h
    Implementation file: MyMemo.cpp

   select the 2nd one in the list
    Edit to
    Class name:          CMyMemoMon
    Header name:         MyMemoMon.h
    Implementation file: MyMemoMon.cpp

   select the 3rd one in the list
    Edit to
    Class name:          CMyMemoDTLinkConverter
    Header name:         MyMemoLink.h
    Implementation file: MyMemoLink.cpp

   select the 4th one in the list
    Edit to
    Class name:          CMyMemoTable
    Header name:         MyMemoTable.h
    Implementation file: MyMemoTable.cpp

   select the 5th one in the list
    Edit to
    Class name:          CMyMemoRecord
    --- it will go to here ----------------
    Header name:         MyMemoTable.h
    Implementation file: MyMemoTable.cpp
    ---------------------------------------
Finish
```

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New Project Information
OK

Developer Studio
Edit -> Find in Files...
Find In Files
  Find what: MymemoRecord
  In files/file types: *.*
  In folder: c:\cdk\samples\mymemo
  check Match case
  check Look in subfolders
  Find

select Find in Files 1 tab
double click on each
  MyMemoLink.cpp
  MyMemoMon.cpp
  MyMemoMon.h
  MyMemoTable.cpp

and perform
Edit -> Replace...
Replace
  Find what:    MymemoRecord
  Replace with: MyMemoRecord
  check Match case
  Find Next
  Replace
  Close
until all are replaced

File -> Save All

==================================================
To Configure a Project to Build a Debug DLL
==================================================
Developer Studio
Project -> Settings...
Project Settings
  Settings For: Win32 Debug
  General tab
    Microsoft Foundation Classes:
      Use MFC in a Shared DLL
    Intermediate files:
      Debug
    Output files:
      Debug
  Link tab
    Category: General
      Output file name:
        Debug/mymemo.dll
OK

Build -> Set Active Configuration...
Set Active Project Configuration
  Project configurations:
    mymemo - Win32 Debug
  OK

Build-> Update All Dependencies...
Update All Dependencies
  check mymemo - Win32 Debug
  OK

Build -> Rebuild All
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============================================================
Set up the Desktop to Run a Debug Synchronization Process
============================================================
If running right mouse the HotSync® process icon
select Exit
run regedit.exe
Registry Editor
  rename the Windows Registry key
  HKEY_CURRENT_USER\Software\U.S. Robotics
  by appending some string to it like
  HKEY_CURRENT_USER\Software\U.S. Robotics_save

On the Windows desktop create a shortcut to
c:\cdk\bim\hsm30\debug\hotsyncd.exe
name it "D restore"

select the D restore icon
right mouse
select Properties
D restore Properties
  select Shortcut tab
    to Target: append  -r
  OK

select the icon D restore
copy and paste it
select the icon Copy of D restore
right mouse
select Rename
enter a name "D verbose"
click to the desktop
right mouse
select Properties
D restore Properties
  select Shortcut tab
  to Target: change "r" to "v"
  OK

double click icon "D restore"

left mouse the HotSync process icon
select Exit

double click icon "D verbose"

place an already initialized device in the cradle
press the HotSync® button
Go through the dialogs as required to establish an account and synchronize the data to the
desktop. When complete there will be a users.dat file and a user folder created in the
c:\cdk\bim\hsm30\debug directory. The users data will be in the user folder.

!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
If there is a display of
Debug Assertion Failed
Abort
The Windows Registry was not restored properley check this and redo it.
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!

At this point the desktop is now set up to run a debug synchronization process.
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==================================================
Set up Windows Registry for your Conduit
==================================================
run regedit.exe
rename
HKEY_CURRENT_USER\Software\U.S. Robotics\Pilot Desktop\Component3
to
HKEY_CURRENT_USER\Software\U.S. Robotics\Pilot Desktop\xComponent3

run
c:\cdk\bin\condcfg.exe
Conduit Configuration
  Add...
  Conduit Information
    select Application
    Conduit: c:\cdk\samples\mymemo\debug\MyMemo.dll
    Creator ID: memo
    Directory: mymemo
    File: memopad.dat
    Remote Database: MemoDB
    Name: MyMemo
    OK
  Exit

run regedit.exe
select
HKEY_CURRENT_USER\Software\U.S. Robotics\Pilot Desktop\Application3

==================================================
Set up HotSync.exe to just run the selected conduit.
==================================================
left mouse the HotSync® process icon
select Exit

double click icon "D vervose"

left mouse the HotSync® process icon
select Custom...
Custom
  double click each listed item except MyMemo
  Change HotSync Action
    select Do nothing
    check Set as default
    OK
  After looping through the list select
  Done

left mouse the HotSync® process icon
select Exit

==========================================
Set up project to run debug DLL
==========================================
Developer Studio
Project -> Settings...
Project Settings
  Settings For: Win32 Debug
  Debug tab
    Category: General
    Executables for debug session:
      c:\cdk\bin\hsm30\debug\hotsyncd.exe
    Working directory:
      c:\cdk\bin\hsm30\debug
    Program Arguments:
      -v
  OK
Build -> Set Active Configuration...
  mymemo - Win32 Debug
  OK
```

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==========================================
To run in debug mode.
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Developer Studio
double click MyMemo.cpp
set a breakpoint in OpenConduit

Make sure the HotSync® process is not running
Build -> Start Debug -> Go
After a short time the HotSync® icon will appear

!!!!!!!!!!!! Improper way !!!!!!!!!!!!!!!!!!!!!!!!!
  Build -> Execute hoysyncd.exe
    The HotSync® icon will appear.
  Build -> Start Debuging -> Go
  Debug output shows hotsyncd.exe exited with code 0x0
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!

==========================================
Set up the device to keep it awake and active.
==========================================
Open the Find dialog, this is not required but allows
you to see the strokes that you are making.

The Graffiti stroke to make is contained between < > symbols.
The Graffiti "shortcut" stroke is a cursive lower case l.
These sequences are 4 Graffiti strokes producing 3 characters.

<The Graffiti shortcuts command stroke, a cursive lower case l> <tap> <tap> <3>

Displays:
No Auto-Off
OK
Stop
Cancel

Select Applications
Select HotSync® icon

PalmPilot -- Continuously press both Up and Down arrow buttons then
             tap the upper right of the screen.

Palm III and after -- Continously press the Up arrow button then
                      tap the upper right of the screen.

Displays:
DLSever Wait Forever is ON
OK

Place the device in the cradle.
Push the HotSync® button

After a short time an arrow will appear pointing to the line where the breakpoint was set.

The code can now be stepped through.

When through select Run to Cursor.

After conduits have run the HotSync.log file is written near the end of the process.

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When Finished
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Do not forget to reset the device so that:
  1) The DLServer will only run when needed.
  2) The unit will not remain active and drain the battery.
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